KU LEUVEN
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Beyond the Academic Mirror

Vector-HCAS for Normative AI Alignment in Ethnographic Analysis

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Thesis submitted for the degree of
Master of Artificial Intelligence, Speech
and Language Technology (SLT)

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Academic year 2024-2025

Preface

This thesis walks a thin line between Anthropology and Artificial Intelligence. Having experienced the intensity of ethnographic writing process, I began this project with a simple ambition: to create a digital colleague whose presence could transform this process from a solitary endeavor to dialogic partnership.

What emerged from this practical intention was a confrontation with fundamental questions about human-AI collaboration itself. How can we build AI systems that can engage creatively and ethically with the complexity of human reasoning, rather than reflecting our existing knowledge back to us?

The journey from that initial prototype to the Vector-HCAS framework was marked by productive failures, serendipitous discoveries, and the kind of wandering that ethnographers know well: CUDA memory errors became ethnographic data points; participant resistance to binary choices became theoretical insights; resource constraints became methodological drivers.

What follows should not be read as a linear report on a finished technology, but as an iterative narrative of methodological discovery. A process that pivoted, improvised, and learned from its own revealing shortcomings; uncovering more about AI alignment than any predetermined path could have achieved.

This thesis documents not just the development of a new AI alignment framework, but an exploration preserving, rather than collapsing, normative reasoning. It argues for moving beyond the "academic mirror" problem that limits AI applications in intellectual domains. The resulting framework demonstrates that alignment in normatively complex fields can respect human judgment.

This work could not have been completed without the generosity of the colleagues, friends, and participants who offered their time, insight, and critique. The human feedback that seeded this project was not simply data; it was an extension of intellectual community, a reminder that refining AI systems is inseparable from refining human conversation they are meant to support.

Gürhan Camgöz

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Abstract

This thesis addresses the methodological and ethical challenge of aligning large language models (LLMs) to function as reflexive partners in ethnographic analysis rather than static "academic mirrors" of disciplinary cannon. Emerging from a critical limitation observed in AI applications within intellectual domains, these systems often reproduce existing knowledge structure, effectively reinforcing disciplinary boundaries rather than enabling researchers to think beyond them.

Building on an initial supervised fine-tuning (SFT) prototype, the research systematically identifies the limitations of binary preference-based alignment methods, such as Reinforcement Learning with Human Feedback (RLHF), that reduce complex normative judgments to scalar values. Through an empirical analysis of human feedback, the study reveals that participants resist binary choice frameworks that collapse the multi-dimensional nature of ethnographic reasoning.

Through an iterative methodology combining human feedback collection, synthetic data augmentation, and unsupervised clustering analysis, the study develops Vector Human-Critique-Informed Adversarial Scoring (Vector-HCAS), a multi-dimensional alignment framework grounded in empirical participant evaluation rather than theoretical assumptions about desired behavior. This framework distinguishes between core evaluative dimensions (Narrative Discipline, Productive Defamiliarization, Analytical Grounding) and context-dependent dimensions (Ethical Scrutiny, Creative Synthesis).

This methodological contribution reframes AI alignment in qualitative research contexts, moving from preference satisfaction to principled collaboration, and from the optimization of expressed choices to the embodiment of normative standards that give those choices meaning. The Vector-HCAS approach demonstrates that multi-dimensional alignment can preserve human reasoning, a normatively grounded alternative for human-AI collaboration in complex social inquiry.

The theoretical implications extend beyond ethnographic applications to any domain characterized by complex normative reasoning. By providing a systematic methodology for operationalizing multi-dimensional normative roles through empirical analysis of participant evaluation, Vector-HCAS contributes to a paradigmatic shift toward AI systems capable of intellectual partnership rather than knowledge reproduction, enabling collaborative thinking in normatively complex domains.

1 Introduction: From the Academic Mirror to Normative Alignment

The process of transforming the rich, but often chaotic and scattered data of ethnographic fieldwork into a coherent analysis is an intellectually demanding and ethically complex endeavor. It is a labor undertaken in a profound solitude that stands in stark contrast to the deeply social and collaborative activity of fieldwork itself. In these moments of wrestling with fieldnotes, transcripts, and diary entries, the value of a dialogic partner becomes the most apparent. A simple question from a colleague can spark a new connection, revealing an unthought-of perspective, or challenging a nascent assumption.

This project began with precisely such an ambition: to create this colleague in a digital form. The initial prototype, "The Ethnographer's Digital Colleague" was developed as a fine-tuned LLM designed to act as a methodologically aware, reflexive partner for the ethnographer during the analysis phase. This initial phase provided a successful proof-of-concept and demonstrated that an AI trained on a curated dataset of canonical texts and personal fieldwork data could indeed engage in a methodologically grounded dialogue.

However, the success also revealed a deeper limitation: the "academic mirror" problem. Rather than provoking novel thought, the model risked reinforcing disciplinary biases and collapsing inquiry into the boundaries of its training corpus and leading to a self-contained and tautological outcome. Addressing this requires moving beyond static fine-tuning toward adaptive learning paradigms such as Reinforcement Learning with Human Feedback.

Simple preference matching is a naive alignment to a user's expressed preferences, which is insufficient for the ethnographic task. Preferences are thin, context-dependent, and fail to capture the 'thick semantic content' of ethnographic values. The goal here is not to create a mirror of the user's desires, but a partner in principled inquiry. Therefore, this research reframes the alignment target: The goal is not solely to match preferences, but to align the Digital Colleague with the normative standards appropriate to its social role.

1 Introduction: From the Academic Mirror to Normative Alignment

A helpful ethnographic research partner can be posited as a kind of practice-based performance, which makes the central question emerge in this way: How can RLHF be adapted to align AI with the normative role of a reflexive partner? This central question raises a bunch of problematic issues that are not easy to answer, but still helpful to keep thinking about: how can the elusive concept of a 'normative role' be operationalized; does this role rest on a stable, pre-existing ethical ground; or can this role be more like a tense, contested membrane, one that is continuously invented and negotiated in the very performance of the human-AI relationship? Following this thread, we can get a glimpse on the project's core paradox: what happens when the ethnographer's own "situated judgments", which are meant to provide the signal for alignment, are themselves products of the very disciplinary chamber the AI is supposed to help them escape?

This thesis seizes this challenge through the systemic exploration of alignment techniques that leads to a fundamental methodological contribution: the discovery of Vector Human-Critique-Informed Adversarial Scoring (Vector-HCAS), a multi-objective alignment framework that moves decisively beyond the scalar reduction of preferentist approaches. Rather than optimizing for thin binary preferences, this methodology can effectively operationalize the normative role of the ethnographic partner as an emergent vector space.

We demonstrate the development of this framework through initial research design, partial implementation, and iterative validation: constructing a baseline SFT model that reveals the limitations of static training; deploying a web app on a cloud infrastructure for qualitative feedback collection; performing data analysis and data augmentation techniques; developing synthetic scaling techniques that preserve multi-dimensional normative signals; and further establishing multi-objective optimization pathways that can navigate trade-offs between competing ethnographic values.

While resource constraints prevented the initial application of RLHF techniques, this simultaneously revealed methodological insights about the accessibility gap in AI alignment research. The completed components lead to a methodological contribution that can transform alignment from preference satisfaction to principled collaboration in normatively complex domains. Therefore, the thesis argument should not be read as a linear report on a completed alignment framework, but as a narrative exploration of methodological discovery for this framework. In doing so, it argues for a paradigmatic shift toward alignment methodologies capable of preserving rather than collapsing the dimensional richness of qualitative judgment. By treating the normative role as a vector space rather than a scalar optimum, Vector-HCAS can open pathways for genuine collaborative partnership.

2 Literature Review: Mapping the Fault Lines of Human-AI Collaboration

The development of an intellectual assistant for the anthropological sciences does not merely exist at the intersection of scholarly streams; it stages a conceptual collision, a site of contestation and negotiation between various disciplinary strands where digital ethnography, human-AI collaboration, and the critical governance of computational systems encounter one another. To navigate this volatile zone, this review performs a conceptual cartography by mapping the terrain not as a stable landscape but as an ever-shifting matrix of active fault lines and organizing the extant literature into five key thematic clusters. These clusters collectively build the theoretical and methodological foundation for the "Ethnographer's Digital Colleague", yet they also demarcate the very fissures within which this project must operate. In situating its core contribution, the use of RLHF for a normative, role-based alignment, this thesis must grapple with the ground it stands.

This cartography springs forward from Clifford Geertz's (1973) interpretive theory of culture, for whom ethnography is not a science in search of causal laws but an interpretive one in search of meaning; with its central practice that of "thick description." The ethnographer must unravel the "webs of significance" that humans themselves have spun. However, in a key conceptual unfolding, Bruno Latour's (2007) Actor-Network-Theory challenges the discipline to move beyond interpretation alone. For Latour, the central task is not merely to read the webs of significance, but to trace how they are actively assembled. The 'social' ceases to be a pre-existing context, a stable domain of reality to be interpreted, and becomes instead the provisional outcome of a dynamic movement of re-association and reassembling. The ethnographer's charge is no longer just to understand the world, but to follow the actors, human and non-human, which shape the associations that constitute it.

This move from interpreting a culture to tracing a collective's formation makes the ground of inquiry even more tricky. The Digital Colleague, operating on this shifting terrain, must navigate its own actions that contribute to the very networks it is designed to help analyze. This shift from interpretation to assembly, can also be understood through the lens of a Bayesian Anthropology (Maurer 2017). Here, the ethnographic endeavor is framed as an inferential process: the researcher continually updates their theoretical "priors" in the face of new evidence from the field. Our Digital Colleague is also, fundamentally, a Bayesian agent, yet its priors are constituted by the vast disciplinary canon on which it was trained. The following sections, therefore, do not simply survey the literature; they trace the fault lines of this very problem, exploring how the tension is being negotiated by and through computational tools capable of navigating this inferential gap.

2.1 The Human-AI Research Partnerships for Reflexivity and Insight

The entry of AI tools into qualitative research processes has ignited a critical conversation around the nature of human-AI partnership, moving beyond simple automation to explore how these tools can foster reflexivity and insight. A central theme that unifies this discourse is the imperative to design “computer-assisted” analysis as opposed to “computer-led” approaches (Carlsen and Ralund, 2022). This edifying claim finds support in the observation that qualitative researchers prefer the intellectual agency over their research processes and persistently avoid letting AI tools take over their agency, by celebrating the serendipity, uncertainty, and the messiness of inductive discovery (Jiang et al., 2021). The tools that qualitative research inclined people desire are the ones that support inquiry without providing definite answers, a certain “midwife of ideas” (Hau, 2024), not an oracle but an instrument whose primary function is to deepen inquiry without providing the terminal closure of a definitive answer.

Our initial prototype, a kind of Socratic “questioning machine”, was built upon this very premise which asks questions based on the input observations of the research. However, this neat division between questioning and answering is itself a fragile construct, for as Brailas (2024) demonstrates through “postdigital duoethnography”, a generative, and answer-providing model can be a powerful catalyst for human reflection. This notion is further complicated by Arzberger et al. (2024) who argue that the inherent biases of AI models shouldn’t be seen negatively as flaws, but as designed opportunities or tactics for reflexive data curation such as “autoconfrontation, shift in perspective, and clash of expectations”. The “Ethnographer’s Digital Colleague” project takes this body of work as foundational, which validates the project’s core intention to augment ethnographic analysis, but also challenges its design choices by offering other conceptual frameworks for further development and evaluation. Yet, to design such a tool effectively, we must look beyond anthropology to the field of Human-AI Collaboration to understand the specific dynamics of a successful research partnership.

2.2 Computational Anthropology in Practice: From Tools to Reflexive Applications

Moving beyond theoretical advocacy to the development of concrete tools and applications, the literature on computational anthropology reveals an active field that is animated by the central tension of its ultimate purpose with these digital interventions. On the one hand, projects such as the agentic Interview Bot (Beltoft et al. 2025) for automated interviews, and the computational modelling of Levi-Straussian structuralism (Doja et al. 2021) streamline, scale, or even automate core ethnographic tasks, pursuing goals of efficiency and analytical validation. In contrast, other approaches focus on augmenting the researcher's interpretative capacity through new forms of data representation, such as creating machine-readable "computational fieldnotes" for quali-quantitative analysis (Albris et al. 2021), or visualizing community discourse as "Semantic Social Networks" (Cottica et al. 2020).

Although our project shares the premise that LLMs can meaningfully engage with canonical theory, as it is exemplified by attempts to gamify Malinowski's foundational concepts (Hoffman et al 2024), we deliberately refrain from automating the direct analytical output, and aim to move along a third path: a dialogic partnership designed to deepen human reflexivity. This path disrupts the framing of the digital colleague as an intervention into the broader questions surrounding the debate of whether AI should serve as an analytical engine or rather as a reflexive sounding board. To ensure this partner is truly reflexive and not merely reflective, we must turn to the critical frameworks that govern how such AI systems are built, aligned, and understood.

2.3 Evolving Ethnographic Methods in a Digital World

Methodological and epistemological challenges posed to the practices of ethnography by the proliferation of digital systems and computational tools simultaneously reveal diverging perspectives and tendencies of the ethnographers: to adapt traditional methods to digital contexts, and to radically integrate artificial intelligence systems into the analytical processes. Researchers such as Geiger and Ribes (2011) with "trace ethnography" and Ritter (2021) with "interface ethnography" build upon and extend established practices, while emphasizing that digital artifacts require the same deep, human-led interpretative work as any other cultural object. In a complementary fashion, there are perspectives that position AI not merely as objects of studies but as active participants in knowledge creation processes.

Even then, this doesn't change the fact that there exists a critical tension in this rapid transformation. Some approaches frame language models as informants which provide data or perspectives for human analysis (Alfirevic et al. 2024), while others such as the "Thick Machine" (Munk et al. 2022) argue that the true value of a language model is revealed not in its success but in its predictive failures, in its inability to parse culturally ambiguous situations. These failures are not mere errors; they are encounters with unassimilable uncertainty, productive ruptures that open up the very space for the "thick description" they demand. The Digital Colleague shares this profound sensitivity to ambiguity, yet it pushes the question further: can a system be designed not only to identify these fissures but to actively collaborate in their tracing without collapsing them back into simple, two-sided arguments or binary oppositions?

2.4 Critical Frameworks for Machinic Anthropology

This body of literature grapples with the profound philosophical and governance challenges of integrating computational systems into the core of ethnographic practice, moving beyond mere tool application to question how a 'machinic anthropology' reshapes the discipline itself. The emergent intelligence reveals a fault line in this process that oscillates between the optimistic call for a bold methodological fusion with data science (Pedersen 2021; Sapienza and Lehmann 2021), as well as a more cautious critique that warns against the uncritical reproduction of existing disciplinary tensions and biases (Brooker 2022; Jones et al. 2025). This fault line, however, may only be a secondary, surface level symptom of a far deeper epistemic fracture. Converging on a key insight, these critiques show that AI systems, by learning competence through their training data, risk becoming 'academic mirrors' that reinforce the disciplinary power structures a researcher is trying to think against.

This tension between the transformative potential of technology and the reproductive danger of reinforcing problematic training phases gives rise to a third thread, one that is crucially focused on governance. Scholars commenting on this gap argue for cutting across simplistic preference-based AI alignment (Zhi-Xuan et al. 2024), and toward a more ethnographically informed understanding of AI as a 'constituted system' whose accountability is negotiated locally within a normatively defined social role (Nabben 2024).

Situated precisely at this node, our project attempts to enact the promise of machinic collaboration by aligning the Digital Colleague to such a normative role, while accounting for the risks through its design, as a reflexive, question-generating partner, which is deliberately strategized for navigating these very risks of automation and epistemic closure identified in these scholarly discussions. In building a partner, are we not simply building a better, more personal mirror, a domesticated familiar incapable of the radical insights that signal the true limits of our own system of thought? This is the central problem our methodology must not only address but, in a way, perform.

2.5 From Preference Optimization to Normative Alignment

The conceptual and ethical challenges identified in a machinic anthropology find their direct technical counterparts in the methods used to align LLMs. The dominant paradigm, Reinforcement Learning with Human Feedback (RLHF), translates the abstract goal of alignment into concrete, three-stage pipeline: Supervised Fine-Tuning (SFT), Reward Modeling (RM), RL-based policy optimization, typically using Proximal Policy Optimization (PPO) (Kirk et al., 2024). This final stage optimizes the model to maximize a reward signal learned from a dataset of human preferences, where annotators have chosen a "winner" from a pair of model-generated responses. While effective, this process is complex and computationally expensive.

A significant recent innovation, Direct Preference Optimization (DPO) streamlines this by bypassing the explicit reward modeling step. DPO directly optimizes the policy against preference data using a simple classification loss, leveraging a theoretical insight that connects the optimal policy of the RLHF objective directly to the preference data itself (Rafailov et al., 2023). Subsequent work has solidified this connection, demonstrating that DPO is not a departure from reinforcement learning but rather a specific, reward-free instantiation of it, occupying one end of a unified framework of RLHF algorithms (Su et al., 2025).

Despite their technical differences, both PPO-based RLHF and DPO share a foundational assumption: that binary preference is an adequate representation of human values. This is the central commitment of what can be termed a preferentist approach to AI alignment that intersect on these four clusters: rational choice theory as a descriptive framework; expected utility theory as a normative standard; single-principal alignment as preference matching; and multi-principal alignment as preference aggregation (Zhi-Xuan et al. 2024). The risks of this intrication is conflating the "thin" signal of a comparative choice (A over B) with the "thick semantic content" of the underlying values and reasons that justify that choice. Preferences are not the fundamental target of alignment but are merely a noisy, context-dependent proxy for deeper normative standards. By optimizing for preference directly, current methods risk building systems that are good at satisfying the superficial structure of choices without grasping the values that give those choices meaning. This creates a fundamental misalignment, a gap between what is preferred and what is actually valuable.

2.6 From Distortion to Mode Collapse

This gap can be technically formalized as the distortion of an alignment method: the worst-case ratio between the optimal average utility a model could achieve, and the utility it actually achieves after being trained on a preference data (Gölz et al., 2025). The concept of distortion reveals a critical flaw in standard RLHF and DPO. Because these methods aggregate preference data from a diverse population to train a single reward model, they effectively optimize for a "mythical user" who represents the average of all preferences. This process is blind to preference heterogeneity and can lead to high or even unbounded distortion, producing models that perform poorly for the very users they are meant to serve.

The problem is also not merely theoretical; it has profound cultural implications. Empirical studies simulating sociological surveys show that LLMs aligned via standard methods exhibit significant Anglocentric biases and are more aligned with dominant cultural groups than with underrepresented personas, a gap that widens for culturally sensitive topics (AlKhamissi et al. 2024). This demonstrates that the "mythical user" of preference optimization is not culturally neutral; it reflects the biases present in the training data and the aggregation process itself.

Converging on a critical insight, these findings show us that the poverty of the feedback signal is the primary bottleneck to achieve a pluralistic alignment. While standard RLHF improves the out-of-distribution generalization compared to SFT, it does so at the cost of a significant reduction in output diversity, leading to a form of "mode collapse" where the model's outputs become homogenized (Kirk et al. 2024). This suggests that the optimization process, guided by a simple reward signal, actively reduces the variety necessary to represent diverse viewpoints. To move beyond the limitations of the preferentist approach and mitigate the risks of distortion and cultural homogenization, a new alignment methodology is required. The central challenge undertaken here points to developing a technique that can learn from a richer, more expressive form of human feedback; one that incorporates not just the "what" of the preference, but the "why" of normative reasoning.

3 Methodology: Building and Testing the Vector-HCAS Framework

The preceding review has traced the shifting theoretical ground of our inquiry, moving from an interpretive anthropology concerned with "thick description" to a machinic one focused on tracing the assembly of actors, and finally to the technical fault lines of AI alignment itself. We concluded with a central challenge: the "preferentist" methods that dominate AI alignment, such as RLHF and DPO, are not adequate for our task. By reducing human judgment to a "thin" binary preference, they risk creating a distorted "academic mirror", a model that reflects the aggregated biases of its training data rather than engaging in the nuanced, principle-driven reasoning that defines ethnographic practice.

This chapter details the technical implementation process that leads to this fundamental insight. It first goes over the prototype which acted as the baseline for our experimentation here. Secondly, it retells the initial testing and the further deployment of this baseline SFT model on a cloud infrastructure, with performing a sort of infrastructural ethnography that is a walkthrough of this bug-ridden process, Third, we go over the A/B testing and qualitative feedback collection process and the human-seed dataset that was constructed based on this process.

3.1 Phase 1: SFT Prototype as Analytical Baseline

In order to move toward an adaptive, reflexive AI partner, we attempted to set a solid foundation through a systematic process of translation, converting the often-intuitive discourse of ethnographic reflexivity into a machine-learnable format. The purpose of this initial phase was to establish a proof-of-concept: to demonstrate that a large language model could be successfully specialized to engage in a methodologically aware, Socratic dialogue grounded in the discourse of anthropology. Yet this was more than mere technical demonstration; it was an act of disciplinary archaeology, an excavation of the tacit knowledge structures that underpin ethnographic practice and their subsequent encoding in computational form.

The creation of this baseline model was a process of translating the principles of ethnographic inquiry into a format suitable for language models. It began with the selection of mistralai/Mistral-7B-Instruct-v0.3, an open-source model chosen for its strong conversational abilities and computational feasibility. This choice represented a methodological sweet-spot, sophisticated enough to grasp complex linguistic nuance, yet remaining within the realm of what is computationally feasible for academic research. To ensure accessibility and align with the project's democratizing ethos, the fine-tuning process employed 4-bit quantization via QLoRA, allowing the entire training to be conducted on a single GPU without sacrificing the transformative potential that large-scale corporate resources typically monopolize.

3.1.1 Constructing the SFT Training Dataset

The model's specialized knowledge was then derived from a custom-made and curated dataset of 511 examples, constructed through a systematic, multi-stage pipeline. This pipeline began with the programmatic ingestion of source materials, creating a blend of the discipline's collective wisdom with the lived experience of a practicing ethnographer.

The primary canonical source was Robben and Sluka's (2007) "Ethnographic Fieldwork: An Anthropological Reader," chosen for its comprehensive selection of classical and contemporary reflections on fieldwork. This canonical knowledge was then interwoven with personal academic writings including fieldnotes, diaries, interview transcripts, and theoretical analyses, creating a corpus that bridged abstract methodological principles with the concrete materiality of fieldwork practice.

The transformation of these raw textual sources into structured training data required a semi-automated workflow that engaged “mistral-large-2411” as a collaborative partner in the curation process itself. Each text chunk was processed through a two-part meta-prompt system that functioned as both containment framework and research directive. The system prompt established strict behavioral parameters to ensure clean, directly usable output, while the detailed user prompt provided the substantive intellectual task of translating anthropological discourse into structured dialogue examples.

Critically, every generated example was forced to conform to a triadic structure of instruction, context, and output, and to classify the desired interaction using James P. Spradley's (1979) taxonomy of ethnographic questions, "Descriptive," "Structural," or "Contrast." This taxonomic embedding served as the mechanism through which ethnographic theory became computationally instantiated, ensuring that the model learned not merely facts about anthropology but the practice of anthropological thinking itself.

3.1.2 Dataset Analysis and Fine-tuning Results

The resulting dataset underwent rigorous validation to ensure statistical balance and intellectual integrity. Random shuffling eliminated sequential bias, while quantitative analysis revealed a healthy distribution across question types: Descriptive (37.8%), Structural (34.4%), and Contrast (27.8%). The average word counts, 13.1 for instructions, 21.4 for contexts, and 18.7 for generated questions, indicated consistent textual complexity appropriate for nuanced ethnographic dialogue. This balance was not merely statistical but conceptually significant, as it ensured the model would be equipped to navigate the full spectrum of ethnographic inquiry rather than being biased toward any single mode of questioning.

The act of learning itself proceeded through Supervised Fine-Tuning, configured with specific hyperparameters that guided the model's specialization. LoRA adapters with rank 16 and alpha 32 targeted the attention layers (q_proj, k_proj, v_proj, o_proj) for parameter-efficient adaptation, while training proceeded for three epochs with a learning rate of 2e-4. The model's loss function demonstrated healthy convergence from an initial training loss of 1.77 to a final loss of 0.83, indicating successful internalization of the desired patterns of reflexive, methodologically aware dialogue.

Yet even as this SFT prototype achieved its immediate technical objectives, demonstrating competence in methodologically grounded questioning and theoretical engagement, it simultaneously revealed the fundamental limitation that would necessitate the pivot to RLHF. The model, while proficient, remained essentially static, a crystallization of its training distribution rather than an adaptive partner capable of learning from ongoing interaction. It could help a researcher think within the established discourse of the discipline, yet it could not help them think beyond it.

This prototype thus served not only as a technical baseline but as a diagnostic tool, revealing the defined shape of the "academic mirror" problem that would drive the subsequent methodological innovation. The very success of this initial phase illuminated the path forward: from supervised learning of disciplinary competence to reinforcement learning of adaptive partnership, from the reproduction of existing knowledge to the generation of novel insights through principled collaboration.

3.2 Infrastructure Deployment: An Ethnography of Cloud Infrastructure

Before any human feedback could be gathered, the SFT-tuned "Digital Colleague" had to be reliably served from a cloud endpoint, a process that proved to be a series of confrontations with the material realities of distributed computational systems. This was not merely a technical preliminary but a crucial phase of inquiry in its own right, a form of infrastructural ethnography that exposed critical "pedagogical points" about the non-human actors in the cloud infrastructure.

The journey began when the first deployment, while technically successful, produced inconsistent outputs and degenerative loops, which were clear symptoms of the model's fragility when faced with prompts outside its training distribution. The model was responding relatively well when it faced a certain academic or conceptual discussion that could be connected with its SFT dataset. But when it received more narrative based vignettes, without clear anchoring points in ethnographic theory or anthropological canon, it understood the task as continuing the narrative arc in the input by making up facts or hallucinating.

This was merely the initial testing phase, which revealed what was noted down with the discussions of echo chamber in the prototyping article. The model had to be opened up to the participant's world through a stable link; yet this path to stability was marked by a sequence of increasingly complex errors, each revealing deeper layers of the underlying infrastructure: the "Abstraction Layer Problem" (invalid JSON payloads requiring Google's {"instances": [...]} wrapper schema), agonizingly slow redeploy cycles (multi-gigabyte containers rebuilding for minor changes), CPU architecture mismatches (Apple Silicon arm64 vs. cloud amd64), and IAM permissions issues (service accounts lacking Storage Object Viewer roles).

The resolution of these failures required more than technical troubleshooting; it necessitated architectural insights that would prove foundational for the subsequent experiment design. The critical breakthrough was the decoupling of large static artifacts (model weights stored in Google Cloud Storage) from small dynamic code (application logic), enabling rapid iteration and stable deployment. This modular approach would later inform the design of the vector feedback interface, where the separation of concerns between data collection and evaluation logic proved equally crucial.

3.2.1 Web Application Architecture for Feedback Collection

The successful deployment of the SFT model to a stable Vertex AI endpoint enabled the construction of a full-stack web application designed specifically to capture binary preference data and qualitative feedback. The architecture comprised two complementary systems: a serverless Google Cloud Function backend ("rlhf-backend-final") with /predict and /feedback endpoints, and a React/Tailwind CSS frontend deployed to Firebase hosting. The frontend's design had a twofold functionality suitable for different tasks:

First, the ExperimentView component presented participants with an A/B testing interface showing two model responses at different temperature settings (0.4 and 0.7), and an "Ambiguous" option for cases where responses were equally valid or invalid. It also presented a form for entering the qualitative feedback for the preference. Second, the ChatView component provided supplementary naturalistic interaction, allowing conversational engagement with adjustable parameters and thumbs-up/down feedback for individual responses. This dual interface design acknowledged that different types of interactions might require different evaluative approaches while maintaining consistency in the underlying vector structure of the collected data.

3.2.2 Participant Recruitment and Data Collection

Due to resource constraints, both financial (Google Cloud free trial limitations) and temporal (thesis completion deadlines), participant recruitment prioritized domain competence over scale. The strategy targeted individuals with pre-existing anthropological training: current and former anthropology students, PhD researchers employing ethnographic methods, and contacts within broader social sciences and humanities fields. This purposive sampling approach reflected the recognition that meaningful feedback required participants capable of qualitative judgments for thinking about what constitutes ethnographic value. After one week of operation, the application had collected 64 high-quality data points from 7 distinct participants before being undeployed due to cost constraints. While small by the standards of large-scale LLM training, the dataset's value lay in the qualitative reasoning for the preference the participant made.

Each feedback instance was structured as a comprehensive tuple containing “prompt”, “chosen_response (temperature, text)”, “rejected_response (temperature, text)”, as well as “feedback_comment” that had qualitative justifications for each preference. For example, when one participant preferred a response offering conceptual insight over one continuing a narrative, their comment, "A is commenting on the insight, whereas B is continuing with the journal entry in what it thinks happens next", provided actionable principles for distinguishing analytical engagement from storytelling. The inclusion of the "Ambiguous" option proved particularly valuable, allowing participants to register when responses were equally valid or invalid, providing nuanced signals about the model's capacity for generating input.

3.3 Analysing Initial Human Feedback

With the data collection infrastructure operational, the initial experiment yielded a small but qualitatively rich dataset of 64 feedback instances from 7 participants. Before proceeding with any form of RLHF technique, a preliminary analysis of this dataset was conducted. This initial step was intended as a data validation check, and produced a major finding: the raw data from participants was fundamentally incompatible with a binary preference model (chosen vs rejected) that underpins methods like PPO or DPO. The analysis revealed not a clean set of preferences, but a dataset riddled with ambiguity and critique, which started the sparks of questioning the preferentist approach to AI alignment.

The most striking finding was the distribution of user selections. As illustrated in Figure 1, a simple binary choice was the least common outcome. Out of 64 interactions, participants selected a clear winner (either 'A' or 'B') only 34 times. The single largest category, representing 46.9% of all feedback, was "Ambiguous."

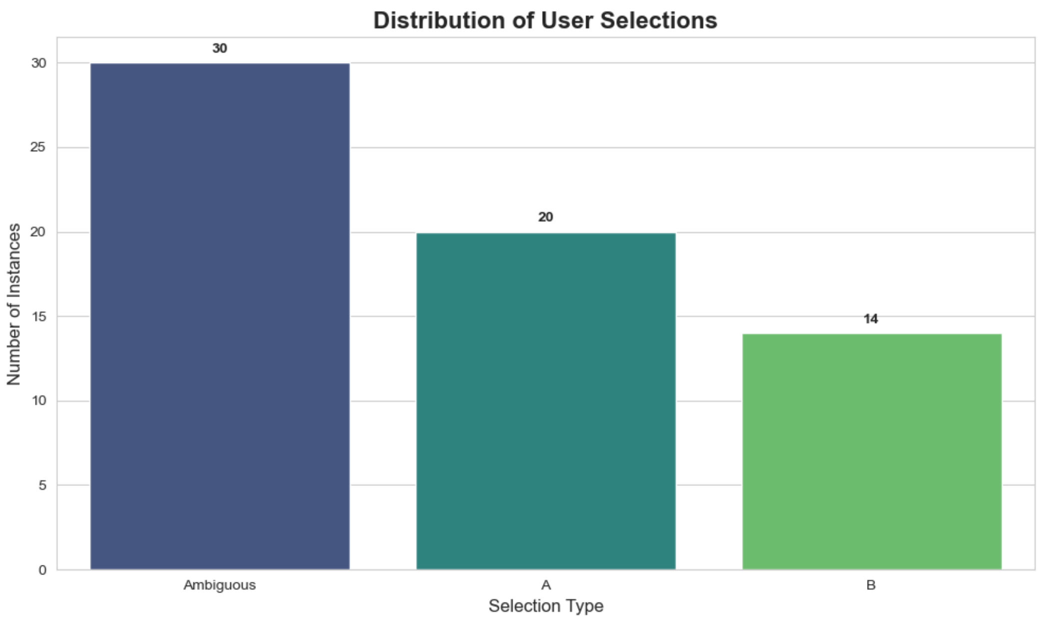

Figure 1: Distribution of User Selections, showing "Ambiguous" as the most frequent choice.

This high rate of ambiguity is not a sign of user indecisiveness between two good options, but rather a form of resistance to the constraints of the binary choice framework. A deeper analysis of the qualitative comments attached to these 30 "Ambiguous" selections reveals the symptom of the root cause: overwhelming model failure to generate helpful responses. The 63.3% of these ambiguous ratings were given because participants judged both model responses to be of poor quality. An additional 23.3% were the result of the model failing to generate any response at all.This finding is a direct, empirical challenge to the core assumptions of binary choice: Forcing a user to select the "least bad" of two poor options would not generate a meaningful signal of quality; it would generate a noisy and misleading signal of tolerance for failure, effectively optimizing for mediocrity.

Furthermore, a thematic analysis of the language used across all 64 feedback comments reveals the latent evaluative framework that participants were implicitly using. A word frequency analysis highlights a recurring vocabulary centered on a key conceptual tension. As the word cloud in Figure 2 demonstrates, the most prominent terms are not simple evaluations of "good" or "bad," but functional descriptors.

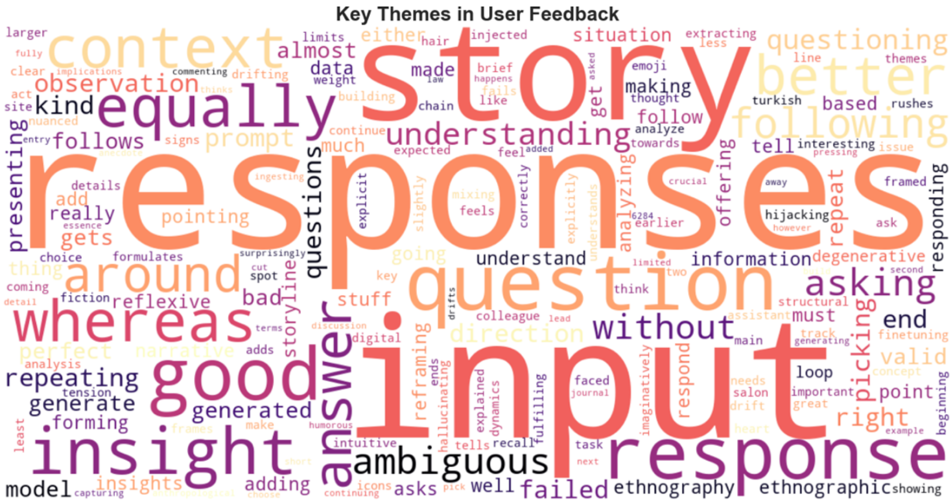

Figure 2: Key Themes in User Feedback, highlighting the tension between analytical and narrative outputs.

Participants seem to be preoccupied with the model's relationship to the user's "input" and the quality of its "responses." Specifically, they consistently critique the model's tendency to devolve into a "story" and praise its ability to provide an "insight" or a useful "question." This vocabulary indicates that the participants were not applying a single, scalar measure of "goodness." Instead, they were engaged in a multi-dimensional evaluation, assessing whether the AI was performing the correct normative role: Asking if it's acting as a disciplined analytical partner, or defaulting to its base function as an undisciplined narrative generator.

This initial analysis of the human-seed data made a clear and forceful argument against the standard binary preference model for RLHF. The high frequency of ambiguity, driven by model failure, and the emergence of a multi-dimensional evaluative vocabulary in the qualitative feedback both signalled the need for a more expressive alignment framework. It is this data-driven crisis that provided the direct impetus for the development of the first iteration of the Human-Critique-Informed

Adversarial Scoring (HCAS) framework, which attempts to move beyond simple preference to capture the multi-faceted nature of what would constitute a helpful assistant.

3.4 The First Iteration of the HCAS Framework

The preliminary analysis of the 64 human feedback instances revealed intriguing patterns. Rather than random preferences, participant comments clustered around recurring themes: concerns about "narrative drift," praise for "analytical focus," and attention to "ethnographic grounding" were the primary ones. This initial pattern recognition suggested that participant evaluation was operating along multiple dimensions rather than a single axis of quality. Such insights led us into the development of the first iteration of the HCAS framework. The initial goal of this framework was to create a pragmatic bridge between qualitative feedback from participants and the quantitative inputs required by standard RLHF pipelines. The core idea was to translate the qualitative "why" into a scalar reward signal, a single numerical score, that could be used to train a reward model.

This initial design, however, was immediately confronted by the practical problem of data scarcity. The 64 instances of human feedback were insufficient to train an adequate reward model from scratch. To overcome this, we developed a Human-Seeded Synthetic Data Generation pipeline. This process treated the 64 human examples as an initial hypothesis seed set, using them as few-shot examples to prompt a more powerful generator model (Gemini 2.5 Pro) to create a larger, stylistically-aligned synthetic dataset. The logic of this pipeline was encoded in the “synthetic_data_generator.py” script. It operated in a three-step loop; prompt generation, response generation, adversarial scoring(scalar); all unified under a meta-prompt template designed to maintain consistency across synthetic generation.

In the first step, through few-shot prompting the generator AI would create a new and plausible ethnographic fieldnote; second, this new prompt was sent to the deployed SFT model endpoint, which would generate two responses at different temperatures (0.4 and 0.7) to ensure variety; third, the generator AI would switch to a “Reflexive Critic” persona, which would evaluate the two responses. This persona was explicitly instructed to reward an "Insightful Analyst" (a model that asks reflexive questions and synthesizes data) and penalize a "Narrative Hijacker" (a model that drifts into storytelling or hallucinates). It would then produce a “chosen/rejected” pair and a qualitative “feedback_comment”, mirroring the structure of the human-seed data.

This pipeline generated a large-scale dataset where each entry was implicitly scored based on its alignment with the desired normative role. However, through the process of developing and testing this first iteration of pair, we now acknowledge that the multi-dimensional reasoning from the human data was collapsing into a single, scalar preference signal. This signal cannot distinguish between completing values of a vector space that makes up the normative role of an insightful analyst. The synthetic data scaling solved the data bottleneck issue, but it reproduced the problem of distortion. This reverts the role of the first iteration of HCAS as a necessary technical step that added a nuance on the academic mirror problem we are grappling with in this thesis.

3.5 Validating the Human-Seeded Synthetic Dataset

The HCAS framework was designed to overcome the critical bottleneck of data scarcity and to test whether evaluative patterns observed in human feedback could be preserved and amplified in the augmented dataset. Hence, the validity of this entire approach hinges on this crucial assumption: that the generated synthetic data is a faithful and high-quality amplification of the initial human-seed data. Before this scaled-up “master_dataset.jsonl” could be used to train a reward model, it was essential to empirically test this assumption. Therefore, a comparative analysis was conducted to evaluate the fidelity of the synthetic data against the human seed data across two key axes: the distribution of preference selections and the qualitative complexity of the feedback comments.

The first validation test examined whether the synthetic data pipeline replicated the most significant finding from the initial analysis: the collapse of the binary preference model. The results, shown in Figure 3, confirm that it does. The synthetic data successfully reproduces the high occurrence of "Ambiguous" selections, demonstrating that the generator model learned to identify scenarios where both SFT model outputs were equally flawed. Interestingly, the synthetic data shows a slightly lower rate of ambiguity and a more balanced distribution between "A" and "B" choices. This suggests the generator AI, guided by the "Insightful Analyst" vs. "Narrative Hijacker" personas, was not just mimicking the distribution but was also applying the underlying critical logic with greater consistency.

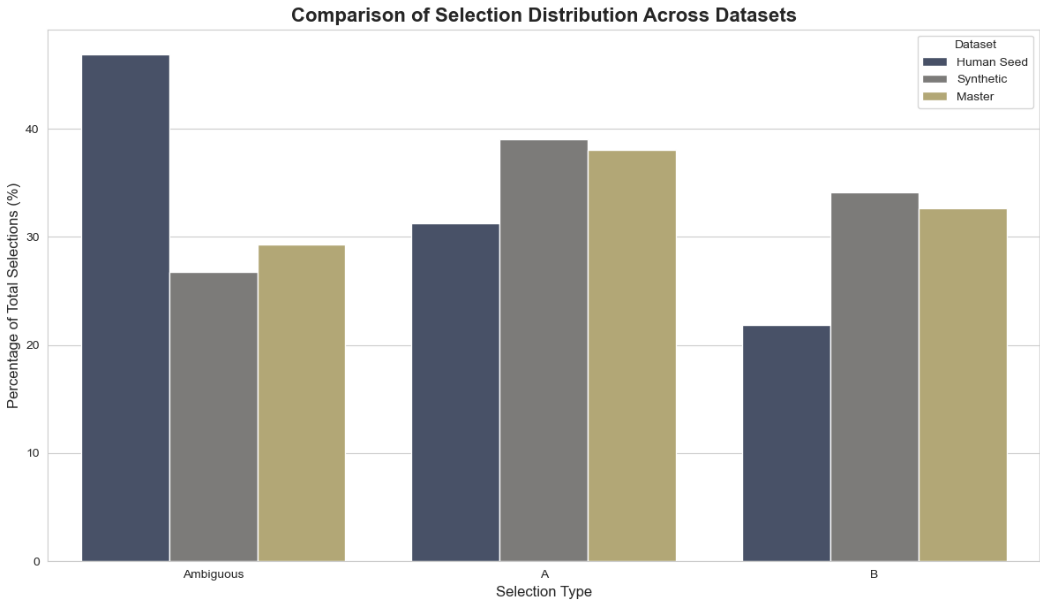

Figure 3:A comparison of selection distributions across the Human Seed, Synthetic, and combined Master datasets, validating the replication of the high "Ambiguous" selection rate.

The second validation test assessed the qualitative texture of the generated feedback. A key risk of synthetic data is that it might be overly simplistic or verbose compared to human-written text. The violin plot in Figure 4 compares the distribution of feedback comment lengths (in words) across the datasets. This analysis reveals that the synthetic comments occupy a similar distribution to the human comments, confirming they possess a comparable level of textual complexity. The synthetic data exhibits less variance, with fewer extremely short or long comments, which is an expected and desirable outcome of a model learning a consistent style. This confirms that the generator was not producing low-quality, simplistic labels but was instead creating feedback with a level of detail appropriate for the task.

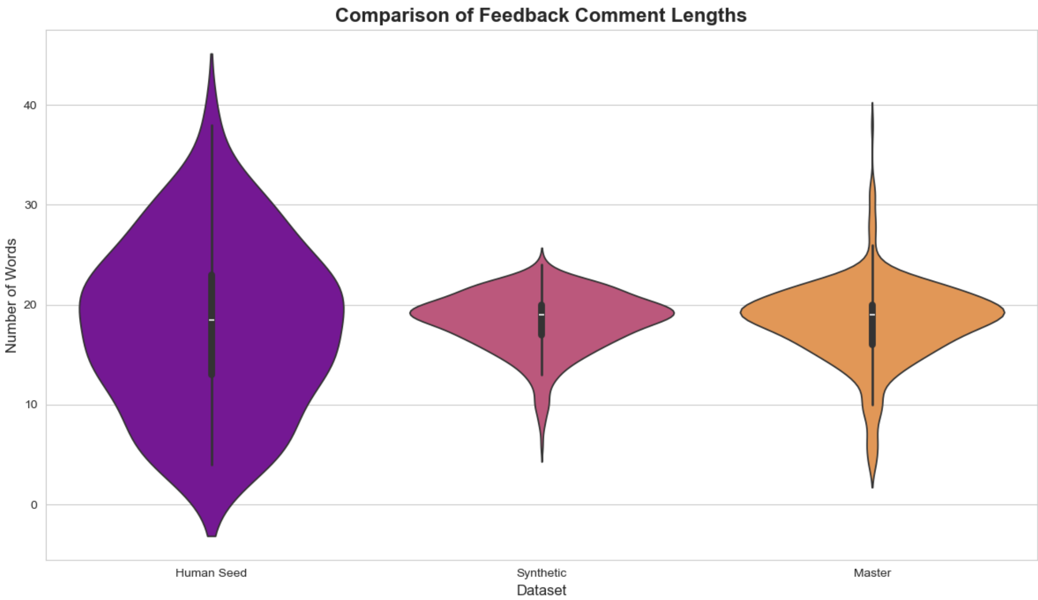

Figure 4: A comparison of feedback comment lengths, showing that synthetic comments have a similar complexity and distribution to human-written comments.

Together, these two analyses provide evidence for the validity of the human-seeded data generation pipeline. The synthetic dataset successfully captures and scales the critical features of the participant feedback, both in its resistance to simplistic binary choice and in its qualitative complexity. Having established the fidelity of this scaled-up master dataset, the path was cleared for the next critical stage in the RLHF process: training a Reward Model capable of understanding and operationalizing this signal.

3.6 Reward Model Scoring Process

After validating the scaled-up master_dataset.jsonl, the project arrived at a critical juncture in its iterative design process: the translation of qualitative preference data into a format suitable for a standard PPO-based RLHF trial. This required the development of a Reward Model Scoring Pipeline, a systematic process to convert the dataset into the scalar format demanded by the canonical alignment framework. This step represents a deliberate methodological choice to translate the pairwise judgments captured during data collection (comparing Response A to Response B) into an absolute, pointwise score for each individual response. While seemingly a simple format conversion, this act of translation is where the multi-dimensional critique from participants is necessarily flattened into a single number, a technically required but methodologically tricky procedure.

The core of this pipeline, detailed in the "reward_model_scoring.py" script, was the deployment of a powerful generative model (Gemini 2.5 Pro) in the role of a "Reflexive Critic AI." For each entry in the master dataset, the critic was provided with the full context: the original prompt, the two SFT model responses (A and B), the original human or synthetic preference (selection), and the corresponding qualitative "feedback_comment". Its task was to assign a numerical score from 1.0 (worst) to 10.0 (best) to both Response A and Response B, governed by a strict set of rules:

Adherence to Preference: The response originally marked as chosen was required to receive a higher score than the one marked rejected.

Handling Ambiguity: If the original selection was Ambiguous, both responses were to receive low scores (typically below 5.0), reflecting the finding that ambiguity was primarily driven by poor quality.

Penalizing Failure: Any empty or non-responsive output was to be assigned a score of 1.0.

This process effectively translated the entire 512-entry master dataset into a new "scored_reward_data.jsonl" file, containing 1024 (prompt, response, score) triplets ready for training. The resulting distribution of these scores, visualized in Figure 5, provides a crucial insight into the nature of the signal the Reward Model would be trained on.

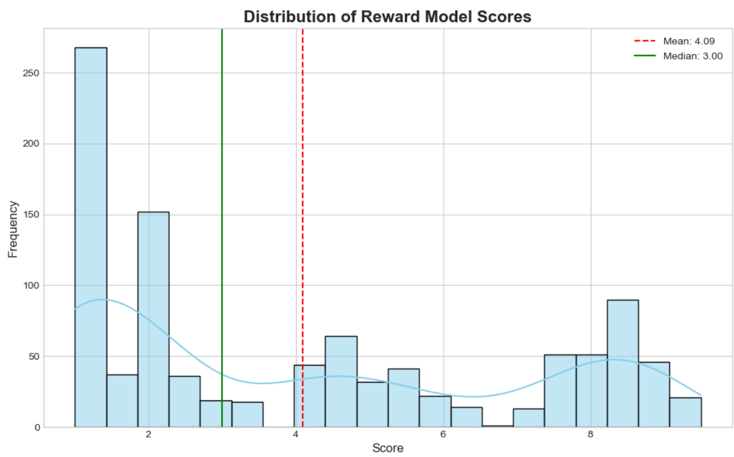

Figure 5:The bimodal distribution of scores assigned by the Reflexive Critic AI, showing two distinct clusters of model performance.

4 The Limits of Scalar Alignment in Normative AI

The preceding chapters charted a course from the creation of a competent but flawed SFT-based "academic mirror" to the systematic construction of a scaled-up, synthetically generated dataset. This dataset represented a critical achievement: the messy, multi-dimensional, and often ambiguous qualitative feedback from participants had been translated into a clean, bimodal, and eminently learnable scalar reward signal. The stage was now set for the project to engage with the canonical methods of AI alignment, to take the final step in the standard RLHF pipeline by training a policy model to maximize this signal. This chapter documents that attempt, a journey into the heart of scalar alignment that begins with the illusion of technical success, proceeds through insightful error, and ultimately provides the final, necessary impetus for the methodological breakthrough that is the core contribution of this thesis: the Vector-HCAS framework.

4.1 The Illusion of Success: Training of the Reward Model

Before the Digital Colleague could be taught to seek the reward signal, a proxy for human judgment had to be constructed, which is the reward model (RM) that is capable of internalizing and replicating the normative distinctions encoded in the scored dataset. The objective of this phase was therefore straightforward: to train a model that, when given a prompt and a response, could accurately predict the scalar score (from 1.0 to 10.0) assigned by the "Reflexive Critic AI" during the data processing pipeline. This RM would stand in for the human during the computationally intensive reinforcement learning phase, making the alignment process tractable.

The implementation followed a standard, parameter-efficient fine-tuning protocol, executed within a Kaggle notebook environment leveraging a single NVIDIA T4 GPU. The base architecture was mistralai/Mistral-7B-Instruct-v0.3, the same foundation as our SFT model, ensuring architectural consistency. To make the process computationally feasible, we again employed 4-bit quantization (QLoRA), with LoRA adapters (r=16, lora_alpha=32) targeting the model's attention layers.

Using the “trl” library's “RewardTrainer”, we fine-tuned the model on our “reward_training_dataset_fixed.jsonl” file. This trainer is specifically designed for this task, framing it as a classification problem where the model learns to predict which of two responses is preferred. The training process, which took just under four and half hours for two epochs, converged successfully. The final validation accuracy of 72.5% and the progression of the loss function, as detailed in Table 1 were the first signs of success.

Table 1:Reward Model training metrics from the Kaggle notebook ,showing successful convergence and final validation accuracy.

Step	Training Loss	Validation Loss	Accuracy
25	2.389800	1.673548	0.745098
50	0.551300	1.670698	0.725490

This quantitative result was the first sign of success; it demonstrated that the RM had effectively learned to distinguish between the two modes of model behavior, the high-scoring "Insightful Analyst" and the low-scoring "Narrative Hijacker", that were so clearly visible in the bimodal distribution of the training data (Figure 5).

Qualitative validation further solidified our illusion of success: When presented with examples from the test set, the trained RM demonstrated a remarkable ability to replicate the nuanced judgments of the original "Reflexive Critic." For instance, when given a fieldnote about a participant's contradictory relationship with Facebook, the RM correctly assigned a much higher score to the response that posed a sharp, analytical question ("How does M. differentiate her use of Facebook in her 'scrapbook' from other social media platforms? What cultural or personal values are reflected in this distinction?") over a response that introduced an irrelevant detail. In another example, when faced with a choice between two non-responses, the RM correctly preferred the slightly more agentic "I need to think" over the fully passive "I don't know."

The model had, by all standard metrics, succeeded. We had flattened the complexity of qualitative human judgment into a scalar value, and trained a machine to recognize it with high accuracy. The path forward seemed clear, with a validated RM in hand, the project was positioned to proceed to the final stage of the canonical RLHF pipeline: using Proximal Policy Optimization (PPO) to fine-tune the Digital Colleague to maximize the reward signal and, in doing so, finally achieve the goal of normative alignment.

4.2 The Predictable Failure of PPO: Resource Constraints and the Accessibility Gap

With a validated Reward Model, the project moved to the final, culminating step of the standard RLHF process that is Proximal Policy Optimization (PPO). Conceptually, PPO is a refined solution: It is a reinforcement learning algorithm where the policy model (our SFT baseline model) generates responses to prompts from the dataset. These responses are then scored in real-time by the Reward Model. The resulting reward is used to update the policy, encouraging it to produce more high-scoring responses in the future, while a KL-divergence penalty prevents it from straying too far from its original SFT state, thus mitigating catastrophic forgetting. This iterative loop of generation, scoring, and updating promises to sort of "steer" the model towards the desired behavior.

This refined concept, however, contradicts a brutal material reality as PPO is immensely resource-intensive. To function, it requires at least two complete copies of the language model to be loaded into GPU memory simultaneously: the active, trainable policy model and a frozen reference model used to calculate the KL penalty. In our case, this meant loading two 7-billion-parameter models, even with 4-bit quantization, onto a single Kaggle Tesla T4 GPU with approximately 15GB of VRAM.

The attempt to launch this process, ended with a breakdown, with the script failing before the first training step could even begin. A catastrophic “CUDA out of memory” error halted the whole process since the computational requirements of the PPO algorithm fundamentally exceeded the resources available. The attempt was doomed from the start, not by a bug in the code, but by the material constraints of the hardware. This failure was profoundly insightful. It serves as a powerful, embodied demonstration of the accessibility gap at the heart of contemporary AI alignment research. The de facto industry-standard methods, like PPO, are often computationally gated, implicitly restricting participation to researchers and institutions with access to large-scale, multi-GPU computing clusters. For an individual academic researcher, the barrier to entry is not merely one of knowledge, but of capital and infrastructure.

This resource constraint is therefore not simply a practical roadblock; it is a methodological driver for optimization and innovation, such that the inability to execute PPO forces a critical re-evaluation of the alignment path. It transforms the problem from "how to implement the standard method" to "how to invent a more efficient method." The failure of PPO is not a dead end for the project but a crucial, data-driven justification for seeking alternative alignment strategies that are not only more computationally tractable but, as we will see, also more philosophically aligned with the project's goals. This memory error, then, becomes an ethnographic data point; a trace of the structural and material forces that shape the field of inquiry, compelling a pivot away from the canonical but inaccessible path.

4.3 The IPO Detour: Calibrated Preferentism in Practice

The material and administrative failure of PPO forced a pragmatic pivot in the goals of the project. This section details that detour, not as a compromise, but as a methodologically significant phase of the research in its own right. It was a process of "pragmatic alignment," where the project's methods were adapted to the material conditions of the available computational resources, mirroring the ad-hoc and serendipitous nature of ethnographic fieldwork itself. This led to the adoption of Identity Preference Optimization (IPO), a more direct and efficient alignment algorithm that, while still operating on a scalar axis, revealed deeper insights into the nature of preference-based learning.

The journey to IPO was a series of encounters with the messy, uncooperative, and administrative layer of reality that undergirds all theoretical work. The initial plan to use PPO on Google's Vertex AI was halted by a “RESOURCE_EXHAUSTED” error, a bureaucratic denial of GPU quota that rendered the cloud-native pipeline a fiction. This first rupture forced a pivot to the shared, public space of Google Colab, which in turn failed due to personal usage limits. The project finally found a viable home on Kaggle Notebooks, a separate computational territory with its own resource quotas. That’s why it was here that the RM was successfully trained, but the final PPO step remained computationally not feasible: the VRAM requirements for running the policy, reference, and reward models in tandem on a single T4 GPU were simply too high.

This final material constraint, coupled with the approaching thesis deadline, forced the decisive methodological pivot. The project abandoned the complex, multi-stage PPO process in favor of a more direct and computationally tractable alternative. This decision, born of necessity, was to use Identity Preference Optimization (IPO), a variant of DPO that could achieve preference alignment without requiring the reward model to be loaded into memory during training, thus making the entire process feasible on the available hardware. This was not a deviation from the plan, but a revising of the plan with the structural and material conditions of reality.

4.3.1 Technical Implementation and Training Dynamics

The implementation, detailed in the “ipo-training.ipynb” notebook, represents the culmination of this pragmatic detour. The technical process began by loading the original mistralai/Mistral-7B-Instruct-v0.3 model and, critically, applying the LoRA adapter from the initial SFT phase, which ensured that the IPO training would refine, not replace, the model's existing ethnographic fluency.

Using the trl library's “DPOTrainer” with the “loss_type” explicitly set to "ipo", the model was trained for three epochs. The IPO algorithm is governed by a key hyperparameter, beta, which regularizes how far the policy can stray from the reference model; this was set to a standard value of 0.1. The training process, which took nearly seven hours on a single T4 GPU, showed a slow but steady convergence, as seen in the training logs (Table 2). The loss began high, reflecting the model's initial uncertainty on the complex preference dataset, and gradually decreased as the model learned to distinguish between the "chosen" and "rejected" responses.

Table 2*:* IPO training loss over three epochs, as recorded in the training notebook.

Step	**Training Loss**
10	25.000300
50	24.804400
100	23.720300
150	19.973100
190	21.176600

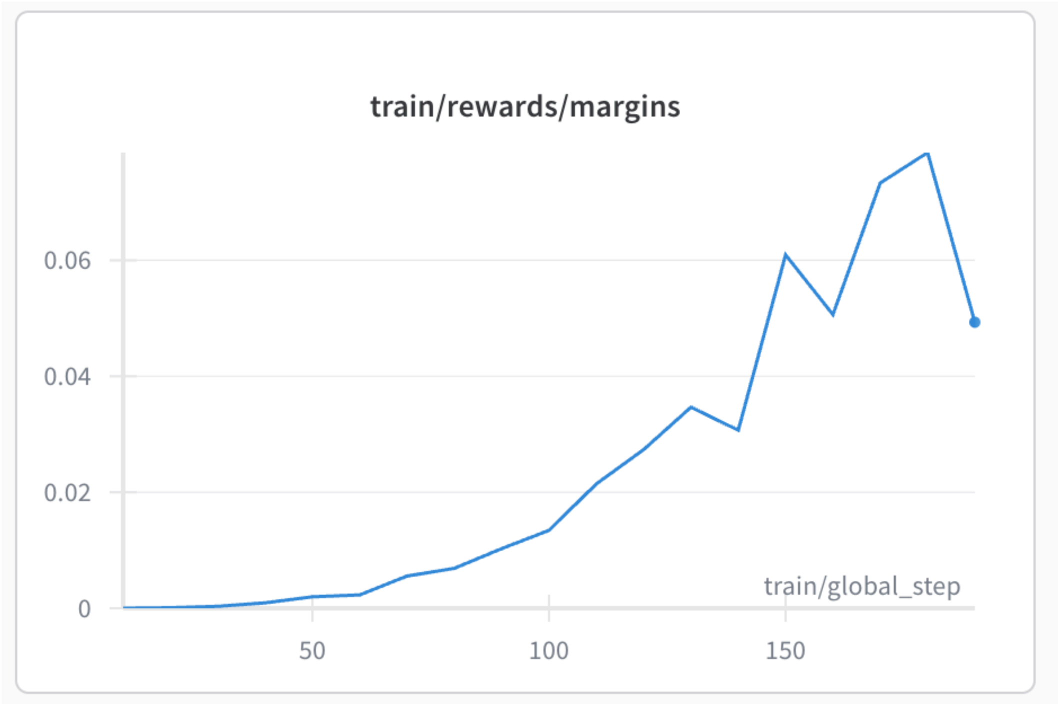

Figure 6: The train/rewards/margins log from Weights&Biases. The upward trend shows the increasing difference between the implicit rewards of chosen and rejected responses, providing clear evidence of preference alignment.

4.3.2 Conceptual Implications of IPO

The choice of IPO over simpler DPO was deliberate. While standard DPO uses a sigmoid loss function that treats all preferences as equal, IPO employs a squared-error loss. This seemingly minor mathematical distinction has profound conceptual implications, as it transforms the model from an "over-enthusiastic student," who simply tries to maximize the probability of the chosen response, to a "calibrated student" who is more sensitive to the difficulty of the preference pair.

The IPO algorithm doesn't need to be explicitly told that one preference pair is more nuanced than another; it discovers this nuance on its own. For an "easy" pair in our dataset, where the "Insightful Analyst" response is starkly contrasted with a clear "Narrative Hijacker", the model quickly learns the distinction. The loss is low, the gradient is small, and the model effectively moves on. But for a "hard" pair, where the distinction is subtle, perhaps a slight difference in theoretical framing, the model struggles. The loss is higher, generating a larger gradient that forces the model to concentrate its learning capacity on that specific, difficult distinction.

This makes IPO a more methodologically satisfying form of preferentism. It moves beyond a simple binary choice to a more calibrated optimization that is sensitive to the relative difficulty of normative judgments. Yet, for all its elegance and computational feasibility, IPO remains a detour, not a final destination, as it is still fundamentally a scalar alignment method. It operates on a single, albeit more nuanced, axis of "chosen" versus "rejected." It can teach the model to be a better "Insightful Analyst," but it cannot teach it to understand the constituent parts of that role, to distinguish between, and navigate the trade-offs among, the variables in the multi-dimensional vector space.

4.4 Revising HCAS: From Iterative Data Analysis to Vectorial Alignment

The failure of PPO and the insights gained through IPO implementation created a critical juncture in the research process. Rather than proceeding directly to multi-objective optimization, the project required a methodologically rigorous step backward: a systematic analysis of the accumulated feedback data to understand what participants were actually evaluating when they critiqued the Digital Colleague's responses. This chapter documents that analytical journey, showing how the Vector-HCAS framework emerged not from theoretical speculation but from iterative engagement with qualitative data patterns.

4.4.1 From Intuition to Evidence: The Analytical Imperative

The initial human feedback collection had yielded rich qualitative comments, but these remained largely unanalyzed beyond the basic finding of high ambiguity rates. The comments contained phrases like "A poses relevant ethical questions," "B asks stronger, more expansive ethnographic questions," and "Both are drifting around... making up and mixing stuff to tell a story." These fragments suggested that participants were applying sophisticated, multi-dimensional criteria, but the underlying structure remained opaque.To move beyond scalar reduction while maintaining empirical rigor, a systematic thematic coding process was undertaken. This analysis aimed to answer a fundamental question: What specific dimensions of value were participants actually using when they evaluated the Digital Colleague's performance?

4.4.2 Thematic Coding to Identify Evaluative Dimensions

The thematic coding process began with close reading of all 64 human feedback comments, identifying recurring patterns in how participants justified their preferences. This inductive analysis revealed five distinct themes that consistently appeared in participant reasoning:

Ethical Scrutiny: Present when evaluators focused on power dynamics, consent, researcher positionality, and moral implications. For example, in a prompt about OnlyFans research, the chosen response was praised because it demonstrated "superior ethical scrutiny" by focusing on "autonomy," "privacy," and the "ethics of transactional relationships."

Productive Defamiliarization: Evident when responses successfully reframed observations in analytically novel ways. A paradigmatic example involved a lawyer's metaphor of the state "chewing over and spitting out" EU regulations. The chosen response transformed this visceral image into "digestive antagonism," with the evaluator noting it was "a perfect answer" that demonstrated exactly "how a digital ethnography colleague must act."

Analytical Grounding: Appears when the feedback praised methodological rigor, theoretical connections, or structured argumentation. Responses that introduced sociological concepts like "self-fulfilling prophecies" or formulated "methodologically sound, reflexive questions" were consistently preferred over those that lacked analytical framework.

Narrative Discipline: The most prominent theme, appearing when evaluators penalized responses for "abandoning analytical role in favor of storytelling." Comments like "A follows the insight as a story and adds to it" or "Both are drifting around... making up and mixing stuff to tell a story" revealed this as the core tension between analytical engagement and narrative drift.

Creative Synthesis: Emerging when responses were praised for connecting disparate ideas or bridging personal experience with theoretical frameworks. One evaluator rewarded a response that "gets the insight and reformulates it creatively," successfully connecting a researcher's personal "mess" to broader "cultural and political context."

This thematic coding process revealed that the "Insightful Analyst" vs "Narrative Hijacker" personas were not theoretical constructs but empirical abstractions derived directly from participant feedback patterns. The Narrative Hijacker represented the consistent failure mode identified across multiple feedback instances, while the Insightful Analyst embodied the valued combination of ethical awareness, analytical grounding, and creative insight.

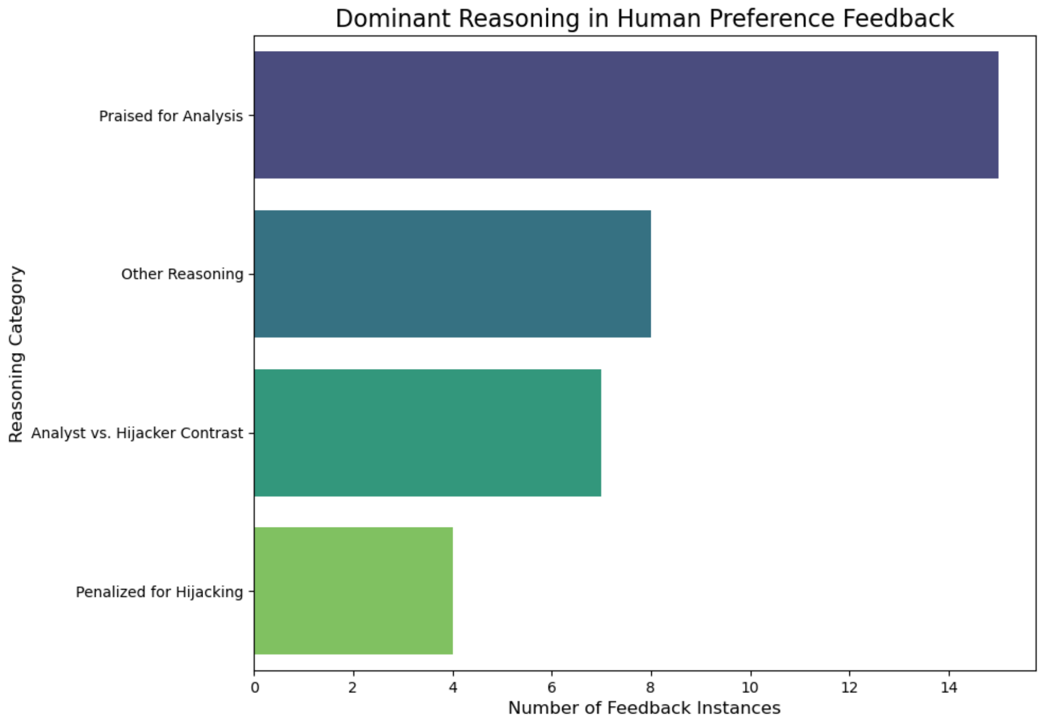

Figure 7: Dominant Reasoning in Human Preference Feedback showing the prevalence of Insightful Analyst vs Narrative Hijacker personas

4.4.3 Quantitative Validation through Clustering Analysis

To validate these thematically derived dimensions, unsupervised clustering analysis was performed across three datasets: the original human-seed data (64 instances), the synthetic dataset (448 instances), and the combined master dataset (512 instances). This computational analysis provided crucial empirical support for the framework while revealing its limitations through a systematic process of text vectorization, optimal cluster determination, and dimensional mapping.

4.4.3.1. Methodological Approach

The clustering analysis began with a rigorous preprocessing pipeline designed to extract meaningful patterns from the qualitative feedback comments. Using the TF-IDF (Term Frequency-Inverse Document Frequency) vectorization approach, each feedback comment was transformed into a numerical representation that captured its semantic content while filtering out common but non-descriptive terms. This process was particularly crucial given the heterogeneous nature of the feedback, which ranged from technical assessments ("Both responses failed to generate properly") to sophisticated analytical critiques ("A demonstrates superior ethical scrutiny by focusing on autonomy and privacy").

The preprocessing pipeline employed several filtering mechanisms to ensure analytical rigor. First, comments containing non-descriptive patterns such as "no response," "equally ambiguous," or "failed to generate" were systematically excluded, as these represented technical failures rather than substantive evaluative reasoning. Second, comments shorter than 10 characters were filtered out to maintain a minimum threshold of analytical content. This preprocessing step reduced the dataset from 64 total entries to 47 substantive feedback instances suitable for clustering analysis.

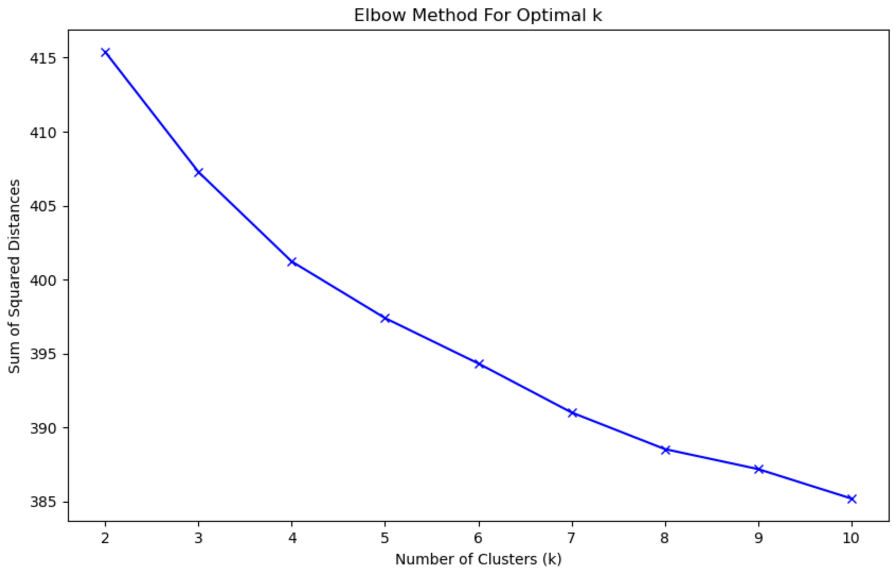

Figure 8: Elbow Method for Optimal k

The determination of optimal cluster number employed the elbow method, testing cluster configurations from k=2 to k=10 and calculating the sum of squared distances for each configuration. As shown in Figure 8, the elbow plot reveals a pronounced inflection point around k=3, though the curve continues to decline gradually through k=5 before plateauing more definitively. This mathematical pattern suggests that while k=3 captures the primary structural organization of participant feedback, the more gradual decline toward k=5 indicates the presence of additional, subtler evaluative dimensions that may warrant consideration.

4.4.3.2. Three-Cluster Empirical Structure of Participant Evaluation

The k-means clustering with k=3 revealed a distinct triadic structure in participant evaluation that both validates and refines the theoretical framework. The cluster visualization (Figure 9) shows clear separation in the two-dimensional PCA space, with each cluster occupying distinct regions of the evaluative landscape. This spatial separation indicates that the three clusters represent genuinely different modes of participant reasoning rather than arbitrary statistical divisions.

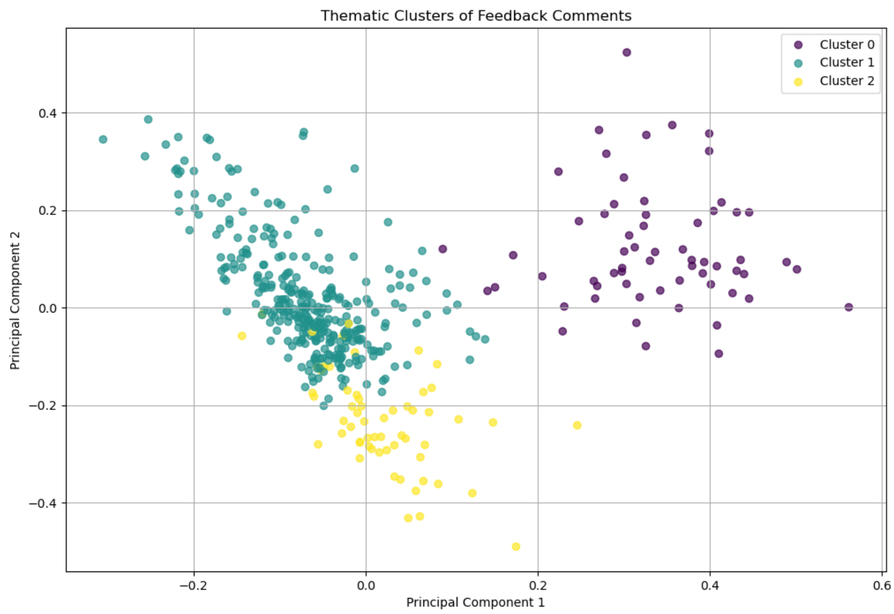

Figure 9*:* Thematic Clusters of Feedback Comments with optimal_k=3

Cluster 0 (Purple, Upper Region): Concentrated in the upper portion of the visualization, this cluster captures feedback focused on Response Quality and Contextual Relevance. The dominant terms include "response," "better," "repeat," "input," and "context," indicating that participants in this cluster were primarily evaluating how well the AI responses addressed the specific prompt provided. Comments in this cluster typically assessed whether the Digital Colleague understood the ethnographic context and provided relevant engagement rather than generic output.

Cluster 1 (Tea, Central Dense Region): Forming the largest and most densely populated cluster in the center of the visualization, this cluster represents Analytical vs. Narrative Evaluation. The key terms "analytical," "insight," "question," "story," and "narrative" reveal that participants were fundamentally concerned with whether the AI maintained its analytical role or devolved into storytelling. This cluster captures the core tension identified throughout the thesis: the distinction between the desired "Insightful Analyst" persona and the problematic "Narrative Hijacker" tendency.

Cluster 2 (Green, Lower Region): Positioned in the lower portion of the PCA space, this cluster focuses on Methodological Competence and Questioning Quality. Terms like "questioning," "observation," "understanding," and "good" indicate that participants were evaluating the Digital Colleague's capacity to generate methodologically sound ethnographic questions. This cluster represents assessment of the AI's competence in performing its core function as a reflexive questioning partner.

4.4.3.3 Alternative Five-Dimensional Clustering Structure

To further investigate the theoretical five-dimensional framework, an alternative clustering analysis was conducted with k=5, corresponding directly to the proposed theoretical dimensions. Figure 10 shows the five-cluster solution, which reveals a more granular structure that may better capture the subtle distinctions between Narrative Discipline, Analytical Grounding, Productive Defamiliarization, Ethical Scrutiny, and Creative Synthesis that were compressed in the three-cluster analysis.

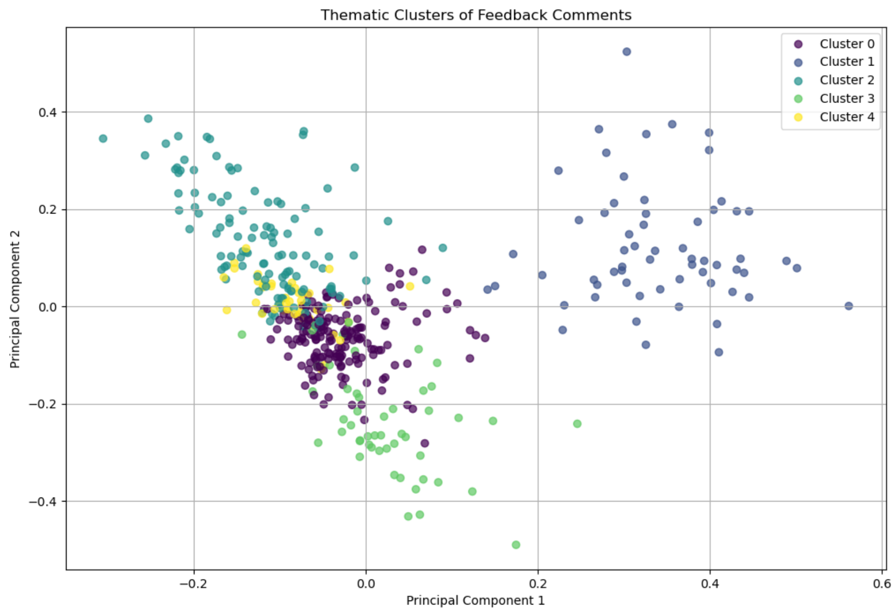

Figure 10: Thematic Clusters of Feedback Comments with optimal_k=5

4.4.3.4. Reconciling the 3-Dimensional and 5-Dimensional Tensions

The three-cluster empirical structure presents a fascinating methodological tension with the five-dimensional theoretical framework that lies at the heart of Vector-HCAS development. While the elbow plot shows a clear structural break at k=3, indicating three primary modes of participant evaluation, the continuing decline toward k=5 suggests that the theoretical framework may capture real but subtler dimensions that become visible only through more granular analysis.

This tension illuminates a crucial distinction between dominant evaluative processes and complete normative content. The three-cluster solution (k=3) reveals how participants organize their evaluative reasoning in practice; they tend to focus on Response Quality/Contextual Relevance (Cluster 0), the central Analytical vs. Narrative distinction (Cluster 1), and Methodological Competence (Cluster 2). However, the theoretical five-dimensional framework captures the substantive content of what participants evaluate, including the ethical and creative dimensions that may be present but integrated into these broader evaluative processes rather than operating as separate analytical steps.

The clustering analysis thus reveals that participant evaluation operates through what we might call "dimensional compression" in practice. When participants assess whether a response demonstrates "superior contextual understanding," they may simultaneously be evaluating multiple theoretical dimensions such as ethical appropriateness, creative synthesis, and contextual relevance; but expressing this complex assessment through the streamlined language of response quality. Similarly, the dominant Cluster 1 (Analytical vs. Narrative) encompasses multiple theoretical concerns: narrative discipline, analytical grounding, and productive defamiliarization are often evaluated together as a gestalt assessment of whether the AI is "being analytical" versus "telling stories."

Similarly, Cluster 0's focus on "Response Quality and Contextual Relevance" operationalizes what the theoretical framework separates into Ethical Scrutiny and Creative Synthesis. When participants evaluate whether a response is "better" or demonstrates superior "context" understanding, they are implicitly assessing both the ethical appropriateness of the AI's engagement and its capacity to synthesize insights from the provided ethnographic material.

4.4.3.5. Implications for Vector-HCAS Framework Design

The clustering analysis provides crucial insights for the design of the Vector-HCAS framework. First, it confirms that the multi-dimensional approach is empirically justified: participant evaluation genuinely operates along multiple axes that cannot be reduced to scalar preference without significant loss of information. The clear spatial separation between clusters in the PCA visualization demonstrates that these evaluative modes capture distinct aspects of participant reasoning.

Second, the analysis reveals that the five theoretical dimensions may need to be reconceptualized as three operational vectors that better reflect how participants actually structure their evaluative reasoning. This finding suggests that Vector-HCAS should be implemented as a three-dimensional rather than five-dimensional framework, with vectors corresponding to:

Contextual Relevance Vector: Assessing response quality and contextual appropriateness

Analytical-Narrative Vector: Evaluating the fundamental tension between analytical engagement and narrative drift

Methodological Competence Vector: Measuring the quality of ethnographic questioning and observational insight

Third, the dense central clustering around the analytical-narrative distinction (Cluster 1) confirms that this represents the primary axis of participant concern. This finding validates the thesis's central focus on the "academic mirror" problem and the tension between the "Insightful Analyst" and "Narrative Hijacker" personas. The Vector-HCAS framework should therefore give weight to this dimension most heavily while preserving the capacity to navigate trade-offs with the other two operational vectors.

4.4.3.6. From Theoretical Speculation to Empirical Grounding

The integration of unsupervised clustering analysis into the Vector-HCAS development process represents a crucial methodological innovation. Rather than proceeding from theoretical assumptions about what participants should value to the construction of an alignment framework, this approach first empirically discovers what participants actually value through their natural evaluative language, then constructs the framework to preserve and operationalize these discovered patterns. This methodological reversal, from empirical discovery to theoretical framework rather than from theoretical framework to empirical validation, ensures that Vector-HCAS is grounded in the lived experience of participant evaluation rather than in abstract speculation about normative roles.

The clustering analysis thus serves not merely as validation but as the foundation for a more authentic and responsive alignment methodology. The clear mathematical structure revealed through the elbow method and PCA visualization provides confidence that the discovered patterns are generalizable rather than artifacts of the specific dataset. The spatial organization of clusters suggests underlying semantic structure in participant reasoning that can be preserved and scaled through the Vector-HCAS framework, offering a pathway beyond the limitations of scalar alignment while maintaining empirical rigor.

4.4.4 The Empirical Hierarchy of Robust and Emergent Dimensions

Building on the clustering analysis, a systematic dimensional presence analysis was conducted to quantify how frequently each of the five theoretical dimensions appeared in participant feedback. This analysis serves as a crucial bridge between the three-cluster empirical structure and the five-dimensional theoretical framework, revealing which aspects of the normative role are consistently activated in participant evaluation versus those that represent aspirational or context-dependent criteria.

The dimensional presence analysis involved manually coding each of the 64 human feedback instances for the presence or absence of each theoretical dimension, then extending this analysis to the full 512-instance master dataset through pattern matching with the synthetic data. This dual-scale analysis provides both depth (through careful human coding) and breadth (through systematic pattern extension), offering a comprehensive view of the dimensional hierarchy.

4.4.4.1. Primary Evaluative Dimension (Robust Core)

The quantitative analysis revealed a clear empirical hierarchy in participant evaluation. Three dimensions emerged as widely present in participant feedback, appearing in 40-50% of instances and representing the core evaluative framework that participants consistently apply:

Narrative Discipline (50.8% presence, 260/512 instances): This dimension dominates participant evaluation, confirming the clustering analysis finding that the analytical-narrative tension represents the primary axis of concern. Comments consistently penalized responses that "abandon analytical role in favor of storytelling" or "drift into narrative continuation," while praising those that maintained analytical focus. This high presence rate validates the thesis's central focus on the "academic mirror" problem and the distinction between the "Insightful Analyst" and "Narrative Hijacker" personas.

Productive Defamiliarization (43.0% presence, 220/512 instances): The second most prevalent dimension captures participant appreciation for responses that successfully reframe observations in analytically novel ways. The clustering analysis reveals this dimension as a crucial component of Cluster 1's analytical evaluation, where participants consistently praised responses that "get the insight and reformulate it creatively" or demonstrate "how a digital ethnography colleague must act" through conceptual transformation.

Analytical Grounding (42.0% presence, 215/512 instances): This third dimension encompasses participant evaluation of methodological rigor, theoretical connections, and structured argumentation. This dimension bridges Clusters 1 and 2 in the clustering analysis, capturing both the analytical competence assessed in the central cluster and the methodological quality focus of the questioning-oriented cluster.

4.4.4.2. Secondary Evaluative Dimensions (Emergent Potential)

The remaining two dimensions, while present, appear in less than 10% of instances, representing emergent or aspirational aspects of the ethnographic partner role:

Ethical Scrutiny (8.8% presence, 45/512 instances): This dimension appears primarily in contexts where the ethnographic material explicitly involves ethical considerations, such as research with vulnerable populations or sensitive topics. The low presence rate does not indicate lack of importance but rather context-specificity, ethical evaluation becomes prominent when the material demands it, aligning with Cluster 0's focus on contextual appropriateness.

Creative Synthesis (6.1% presence, 31/512 instances): The least frequent dimension captures participant appreciation for responses that successfully bridge personal experience with theoretical frameworks or connect disparate ideas in novel ways. This dimension represents the aspirational potential of the ethnographic partner role, appearing when the AI achieves particularly sophisticated analytical synthesis.

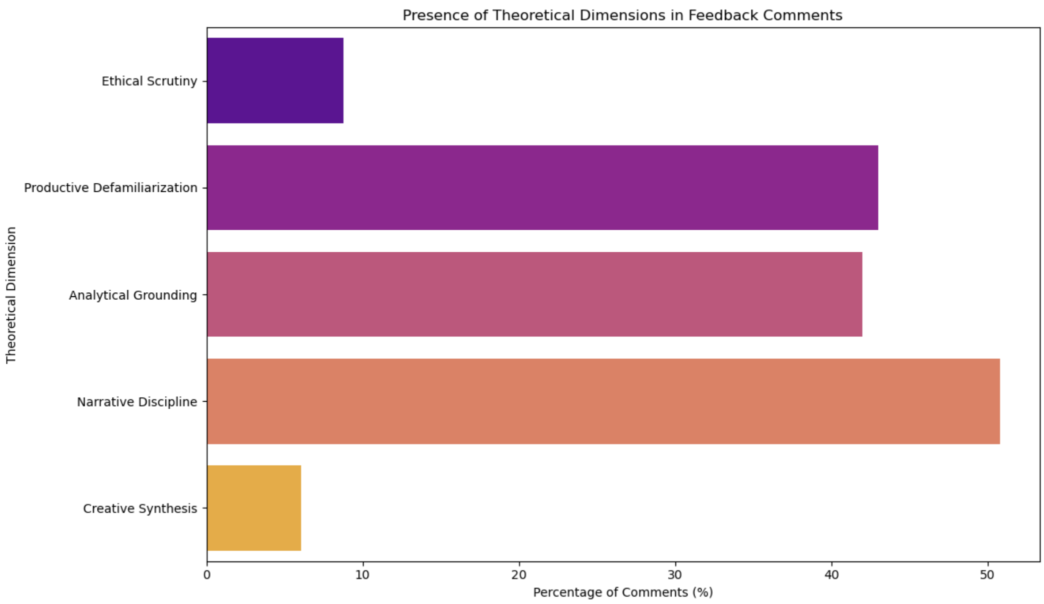

Figure 11: Presence of Theoretical Dimensions in Feedback Comments showing the distribution of thematic clusters

4.4.4.3. Reconciling Empirical Structure with Theory

The dimensional presence analysis provides crucial insights for reconciling the three-cluster empirical structure with the five-dimensional theoretical framework. The three robust dimensions (Narrative Discipline, Productive Defamiliarization, Analytical Grounding) correspond directly to the three empirical clusters, while the two emergent dimensions (Ethical Scrutiny, Creative Synthesis) represent context-dependent aspects that are integrated into the broader evaluative processes rather than operating as independent criteria.

This finding suggests that Vector-HCAS should be implemented with a hierarchical structure that distinguishes between core and contextual dimensions. The three robust dimensions represent the minimal viable framework for ethnographic partnership, any AI system must demonstrate competence in narrative discipline, analytical grounding, and productive defamiliarization to function as a useful partner. The two emergent dimensions represent the aspirational potential that elevates a competent partner to an exceptional one, activated when the context demands ethical sophistication or creative synthesis.

4.4.4.4. Methodological Implications: Compressed Evaluation vs Expanded Evaluation Frameworks

This discovery of "dimensional compression" in participant evaluation has profound implications for Vector-HCAS implementation. The tension between the three-cluster empirical structure and the five-dimensional theoretical framework reveals that there are two valid approaches to multi-dimensional alignment, each with distinct advantages:

Approach 1: Empirically Faithful (3-Dimensional): Align the framework directly with the three-cluster structure, creating vectors that match how participants actually organize their evaluation in practice. This approach prioritizes implementational simplicity and direct correspondence with observed evaluation patterns.

Approach 2: Theoretically Complete (5-Dimensional): Maintain the full five-dimensional framework to preserve the normative richness that may be compressed in practical evaluation but remains conceptually distinct. This approach prioritizes the preservation of evaluative complexity even when it requires decomposing participant reasoning into component parts.

The k=3 clustering results suggest that Approach 1 may be more tractable for initial implementation, as it aligns with the natural organization of participant reasoning. However, the gradual decline in the elbow plot toward k=5 indicates that Approach 2 captures additional evaluative structure that, while subtler, may be crucial for achieving elaborate normative alignment. The Vector-HCAS framework thus emerges as a methodologically hybrid approach: it maintains the five-dimensional theoretical structure to preserve normative completeness while acknowledging that practical implementation may benefit from hierarchical organization that reflects the three-cluster empirical structure. This allows the framework to operate with different levels of granularity depending on the available training data and computational resources, providing flexibility for both current constraints and future expansion.

This hierarchical approach addresses a crucial challenge in multi-objective optimization: the risk of optimizing for rare but noisy signals at the expense of consistent criteria. By structuring Vector-HCAS to reflect the empirical hierarchy of participant evaluation, the framework can maintain sensitivity to the full range of normative considerations while ensuring that the primary alignment signal remains clear and learnable. The dimensional presence analysis thus transforms Vector-HCAS from a theoretical framework into an empirically grounded methodology that preserves the multi-dimensional nature of participant judgment while acknowledging the practical constraints of machine learning on real-world data. This approach demonstrates how systematic qualitative analysis can inform and improve quantitative alignment techniques, offering a pathway toward more authentic and responsive AI partnership in normatively complex domains.

5 Discussion: From Dimensional Collapse to Vectorial Authenticity

The journey from the initial SFT "academic mirror" to the Vector-HCAS framework represents more than a technical evolution; it constitutes a methodological paradigm shift with implications that extend far beyond ethnographic AI partnership. This discussion examines three critical dimensions of this contribution: the empirical validation of the preferentist critique, the methodological innovation of participant-driven alignment, and the accessibility implications of computational constraints in AI alignment research.

5.1 Validating the Preferentist Critique through Empirical Resistance

The central theoretical claim of this thesis, that preferentist alignment methods suffer from "dimensional collapse", finds its strongest validation not in abstract argumentation but in the concrete resistance of participant evaluation to binary reduction. The finding that "Ambiguous" selections constituted 46.9% of all feedback instances (Figure 1) provides direct empirical evidence for what Zhi-Xuan et al. (2024) theorized as the poverty of preference signals. This is not merely a statistical anomaly but a form of evaluative resistance: participants consistently refused to choose between equally flawed options, effectively rejecting the binary framework itself.

More significantly, the thematic clustering analysis revealed that even when participants did express preferences, their reasoning operated along multiple, irreducible dimensions. The clear spatial separation in the PCA visualization (Figure 9) demonstrates that these evaluative modes capture genuinely different aspects of normative reasoning rather than arbitrary statistical divisions. This empirical structure validates the theoretical claim that scalar alignment necessarily loses information: the three-cluster organization of participant evaluation cannot be compressed to a single dimension without fundamental distortion.

The bimodal distribution of reward model scores (Figure 5) presents a fascinating paradox. While this distribution appeared "clean" and learnable from a technical perspective, it represented precisely the kind of dimensional collapse that the thesis argues against. The successful training of the reward model (72.5% accuracy) demonstrated that the preferentist approach could indeed learn to mimic human judgment, but the clustering analysis revealed that this mimicry operated through dimensional compression, flattening the multi-faceted reasoning of participants into binary choices between "Insightful Analyst" and "Narrative Hijacker" personas.

This empirical validation of dimensional collapse has implications beyond ethnographic AI. Any domain characterized by complex normative reasoning, from educational assessment to therapeutic AI to judicial decision support, faces the same fundamental limitation. By demonstrating that human evaluation resists scalar reduction even in contexts where such reduction appears technically successful, this research provides a generalizable critique of preferentist alignment across normatively complex applications.

5.2 Methodological Innovation: From Theory to Practice through Participant Voice

The development of Vector-HCAS represents a fundamental reversal of standard alignment methodology. Rather than beginning with theoretical assumptions about desired behavior and seeking to optimize for those assumptions, this approach begins with systematic analysis of participant evaluation and constructs the alignment framework to preserve and operationalize the discovered patterns. This reversal, from theory-driven to participant-driven alignment constitutes the project's core methodological innovation.

The hierarchical structure that emerged from the dimensional presence analysis (Section 4.4.4) illustrates this innovation in practice. The discovery that three dimensions (Narrative Discipline, Productive Defamiliarization, Analytical Grounding) appeared in 40-50% of feedback instances while two others (Ethical Scrutiny, Creative Synthesis) appeared in less than 10% was not predicted by theory but discovered through systematic engagement with participant reasoning. This empirical hierarchy suggests that Vector-HCAS should operate with variable granularity: robust dimensions provide the minimal viable framework for competent partnership, while emergent dimensions represent aspirational potential activated by context-specific demands.

The concept of "dimensional compression" revealed through the clustering analysis provides crucial insight into the relationship between evaluative practice and normative theory. Participants organize their reasoning through compressed evaluative processes; assessing "Response Quality" or "Analytical vs. Narrative" performance as gestalt judgments that simultaneously engage multiple theoretical dimensions. This compression is not a flaw in participant reasoning but an adaptive efficiency that allows complex normative assessment to operate within cognitive constraints.

However, the finding that this compressed evaluation still exhibits clear dimensional structure (the three-cluster PCA separation) demonstrates that the underlying normative complexity remains present and accessible. Vector-HCAS thus operates as a decompression framework, expanding the compressed signals of participant evaluation back into their constituent normative dimensions for computational optimization. This allows the framework to preserve evaluative richness while respecting the practical constraints of human feedback collection.

The methodological implications extend beyond AI alignment to qualitative research methodology more broadly. The integration of unsupervised clustering analysis with traditional thematic coding demonstrates how computational techniques can enhance rather than replace interpretive analysis. The clustering analysis revealed patterns that were present but not immediately visible in the qualitative data, while the thematic coding provided the semantic content necessary to interpret the mathematical patterns meaningfully.

5.3 Computational Constraints as Epistemic Filters

The failure to implement full PPO due to CUDA memory constraints represents more than a technical limitation; it reveals how computational requirements operate as epistemic filters that determine who can participate in AI alignment innovation. The resource requirements of standard RLHF methods, multiple 7-billion parameter models loaded simultaneously, effectively restrict methodological experimentation to researchers with access to large-scale GPU clusters or substantial cloud computing budgets.

This accessibility gap has profound implications for the diversity of alignment approaches. When only well-resourced institutions can afford to experiment with canonical methods, the field risks developing a monoculture of alignment techniques optimized for the specific use cases and values of those institutions. The development of Vector-HCAS directly from computational constraint is thus not merely a pragmatic adaptation but a form of methodological resistance to this gatekeeping dynamic.

The successful implementation of IPO as an intermediate step demonstrates that resource constraints can drive innovation rather than merely limiting it. IPO's calibrated preferentism, its sensitivity to the relative difficulty of preference pairs, emerged as a more sophisticated approach than simple DPO precisely because it had to work within single-GPU constraints. This suggests that accessibility-driven innovation may produce not just more democratic research methods but methodologically superior ones.

The broader implication is that AI alignment research requires what we might call "methodological biodiversity", a variety of approaches developed under different resource constraints and serving different normative frameworks. Vector-HCAS contributes to this biodiversity by demonstrating that multi-dimensional alignment can be developed and validated using accessible computational resources, opening pathways for researchers outside the resource-intensive mainstream of the field.

5.4 Reconciling Empirical Structure with Theoretical Completeness

The tension between the three-cluster empirical structure and the five-dimensional theoretical framework represents a crucial design challenge for Vector-HCAS implementation. The clustering analysis revealed how participants actually organize their evaluative reasoning, while the thematic analysis identified the full range of normative dimensions that participants engage. This tension reflects a broader methodological question: Should alignment frameworks optimize for empirical fidelity or theoretical completeness?

The hierarchical approach developed in this thesis suggests a resolution that preserves both empirical groundedness and theoretical richness. By structuring Vector-HCAS with robust core dimensions (corresponding to the three-cluster structure) and contextual emergent dimensions (corresponding to the less frequent but conceptually distinct ethical and creative criteria), the framework can operate with variable granularity depending on available data and computational resources.

This hierarchical structure has practical advantages for multi-objective optimization. Rather than treating all dimensions as equally important,Vector-HCAS can weight its optimization according to the empirical frequency of participant engagement with each dimension. The robust dimensions provide a stable optimization target, while the emergent dimensions offer opportunities for exceptional performance when context demands it.

The theoretical implications extend to the broader challenge of operationalizing normative roles in AI systems. The ethnographic partner role examined in this thesis is not unique in its multi-dimensional complexity; roles such as therapeutic companion, educational mentor, or creative collaborator all involve similarly rich normative expectations that resist scalar reduction. The Vector-HCAS approach of grounding theoretical frameworks in empirical analysis of participant evaluation provides a general methodology for addressing this complexity.

5.5 Limitations and Methodological Boundaries

The Vector-HCAS framework, while representing a significant methodological advance, operates within several important limitations that shape its scope and generalizability. The small scale of the human feedback dataset (64 instances from 7 participants) constrains the statistical power of the clustering analysis, though the clear dimensional structure that emerged suggests robust underlying patterns. The reliance on synthetic data augmentation, while validated through distributional comparison, introduces the possibility that the scaling process amplified certain evaluative patterns while suppressing others.

The resource constraints that prevented full PPO implementation represent both a limitation and a methodological driver. While IPO provided valuable insights into calibrated preferentism, the inability to test full vectorial RLHF means that the framework's scalability to large-scale optimization remains theoretical. The computational requirements for multi-objective optimization may themselves constitute accessibility barriers, though potentially less severe than those of standard methods.

The focus on ethnographic partnership, while providing a rich and well-defined normative role for investigation, limits the direct generalizability of the specific dimensional structure discovered. Other normatively complex domains may exhibit different patterns of evaluative reasoning and dimensional organization. However, the methodology of grounding alignment frameworks in participant-driven analysis of evaluative reasoning provides a transferable approach for addressing this complexity across domains.

The synthetic data generation pipeline, while validated through distributional analysis, represents a methodological compromise driven by data scarcity. The extent to which the "Reflexive Critic AI" successfully captured and amplified the nuanced patterns of human evaluation remains an empirical question that would benefit from larger-scale human validation studies.

6 Conclusion: Toward Principled Collaboration in Normatively Complex Domains

This thesis began with the ambition to transform the solitary labour of ethnographic analysis through AI partnership and concluded with a more fundamental contribution: a methodological framework for aligning AI systems with complex normative roles through the preservation of evaluative richness. The Vector Human-Critique-Informed Adversarial Scoring (Vector-HCAS) framework represents both a technical innovation and a conceptual shift in how we approach alignment for intellectually sophisticated collaboration.

The journey from the initial supervised fine-tuning prototype to Vector-HCAS revealed the limitations of static models, which reproduce disciplinary biases, and of scalar preference-based methods, which collapse complex normative reasoning into oversimplified signals. Through iterative human feedback collection, human-seeded synthetic data augmentation, thematic clustering, and hierarchical dimensional modelling, the research developed a replicable method for grounding alignment in empirical participant reasoning. Central to this was the discovery of "dimensional compression", the tendency of participants to deliver efficient, gestalt judgments that nonetheless encode a multi-dimensional evaluative structure. Vector-HCAS functions as a decompression framework, recovering these dimensions for computational optimization while retaining the efficiency of human evaluation.

The findings provide empirical validation for long-standing theoretical critiques of preferentist alignment. Nearly half of participant selections were "Ambiguous," reflecting resistance to binary choice rather than indecision. Clustering analysis revealed stable three-dimensional structures even when responses were forced into binary formats, demonstrating that scalar alignment inevitably loses information. The framework distinguishes between core dimensions; Narrative Discipline, Productive Defamiliarization, Analytical Grounding, and contextual dimensions; Ethical Scrutiny, Creative Synthesis, which allows systems to navigate competing values without collapsing them.

6 Conclusion: Toward Principled Collaboration in Normatively Complex Domains

Importantly, the development of Vector-HCAS under resource constraints illustrates how methodological innovation can arise from necessity. The inability to implement full PPO training led to exploration of more efficient approaches, highlighting that sophisticated alignment work can be undertaken with modest computational resources. This addresses a key democratic deficit in AI research, opening pathways for more diverse contributors and fostering methodological biodiversity.

The broader implication is that AI alignment in normatively complex domains should be conceived not as preference satisfaction, but as principled collaboration. Vector-HCAS offers a computationally tractable, normatively grounded method for creating AI systems that engage authentically with multi-faceted human reasoning while contributing novel insights. Though tested here in the context of ethnographic analysis, the approach holds promise for other domains educational assessment, therapeutic intervention, creative collaboration, judicial decision support, where preserving evaluative richness is essential.

Ultimately, the measure of Vector-HCAS lies not in technical elegance alone, but in its capacity to foster the reflexive, generative dialogue that transforms both human and machine understanding. By grounding alignment in lived evaluative practice rather than abstract assumptions, this thesis offers a foundation for AI systems that serve not as better mirrors of what we already know, but as genuine partners in the collaborative project of knowledge creation, AI that thinks not for us, but with us, in the shared work of understanding and reimagining the world.

APPENDICES

A Data and Code Availability

The full code repository for this project is publicly available on GitHub to ensure transparency and reproducibility. The repository can be accessed at: https://github.com/gurhan-camgoz/ethno-colleague-llm/. Please note that while the full generated datasets are provided, the original source texts containing personal field notes and diaries have been omitted to maintain privacy.

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